

- 3** Five descriptions of different input or output devices are given in the table.

Complete the table by stating the **name** of each input or output device.

Description	Name of device
This is an input device that works by shining a light onto the surface of a document. The light source is automatically moved across the document and the reflected light is captured by mirrors and lenses.	
This is an input device where a laser or a light source is moved across an object. The width, height and depth of the object are measured to allow a model to be created.	
This is a large input device that is usually fixed to a wall. A user can calibrate the device to make sure the sensors align with a projected image. The user can use either their finger or a special pen to make selections.	
This is an output device that uses many small mirrors to reflect light towards a lens. This will display an image.	
This is an output device that creates an object by building layer upon layer of material.	

[5]

- (b)** The law company wants to purchase a new file server.

The company can purchase a server with either solid state storage or magnetic storage. After discussion, it decides to purchase a file server with magnetic storage.

Explain why the company chose magnetic storage rather than solid state storage.

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[4]

- (c) The law company also uses optical storage.

Give **three** different examples of optical storage.

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[3]

0478/12 – JUNE 2019

- 1 Input and output devices are often connected to a personal computer.

- (a) Identify **three** input devices that can be connected to a personal computer.

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[3]

- (b) Identify **three** output devices that can be connected to a personal computer.

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[3]

- 2 A finance company uses off-line storage to archive their accounts.

- (a) Explain what is meant by off-line storage.

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..... [2]

- (b) The computers in the finance company use both primary and secondary storage.

- (i) Give **one** example of primary storage.

..... [1]

- (ii) Give **two** examples of secondary storage.

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[2]

- 3 Vanessa writes a paragraph as an answer to an examination question about the central processing unit (CPU).

Use the list given to complete Vanessa's answer by inserting the correct **six** missing terms. Not all terms will be used.

- Components
- Data
- Decoded
- Executed
- Fetched
- Instructions
- RAM
- ROM
- Secondary storage

The CPU processes and

An instruction is from

into the CPU where it is then Once this has taken place the instruction is then

[6]

- 4 (a) Marley wants to store a video he has created for his school project.

He considers using a DVD or a Blu-ray to store the video.

Explain **two** differences between a DVD and a Blu-ray.

- 1
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- 2
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[2]

(ii) Marley prints the images for his project using an inkjet printer.

Describe how the inkjet printer prints an image.

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..... [4]

7 (a) An office has an automated lighting system. When movement is detected in the office the lights are switched on. If movement is not detected for a period of 2 minutes the lights are switched off. The system uses a sensor and a microprocessor.

Describe how the automated lighting system uses a sensor and a microprocessor.

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..... [6]

(b) A microprocessor uses ROM.

Explain what is meant by ROM.

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..... [3]

10 Remy has a mobile device that has a capacitive touch screen.

Describe how the capacitive touch screen registers Remy's touch.

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..... [4]

- 2 (a) A computer can have both a MAC address and an IP address.

Four statements are given about MAC addresses and IP addresses.

Tick (✓) to show whether each statement is **True** or **False**.

Statement	True (✓)	False (✓)
A MAC address is unique to a computer on a network		
Once an IP address has been set it cannot be changed		
A MAC address is made up of the computer's serial number and the IP address		
If a computer does not have an IP address it cannot communicate with another device using the Internet		

[4]

- (b) A computer uses the Von Neumann model and the stored program concept.

- (i) Explain what is meant by the stored program concept.

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..... [2]

- (ii) The Von Neumann model has several components that are used in the fetch-execute cycle.

One component is the Arithmetic Logic Unit (ALU).

Describe the role of the ALU.

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6 A museum has an information point.

Visitors to the museum can use the information point to plan their visit to the museum.

The information point allows visitors to access the information using a resistive touch screen.

Visitors can either listen to the information or read it on the screen. They can also select to output a paper copy of the information they require.

(a) Describe how the resistive touch screen registers the visitor's touch.

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..... [4]

(b) The information point has a screen to allow visitors to read information.

Identify **two** other output devices that are present in the information point.

Output device 1

Output device 2 [2]

(c) The information point uses both primary and secondary storage.

Explain what is meant by primary and secondary storage.

Primary

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Secondary

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..... [4]

1 Andrew wants to produce advertising material for his company.

(a) Andrew can use an **Inkjet printer** or a **Laser printer**.

Draw lines to connect each printer to a correct statement. More than one line may be used to connect to each printer or statement.

Printer	Statement
	Can print in colour
Inkjet printer	Uses a charged drum to create the printed item
	Uses powdered toner
Laser printer	Creates output line by line using a print head

[2]

(b) Andrew wants to print a single page A4 leaflet. He wants to print 10 000 copies.

State whether he should use an inkjet or a laser printer.

..... [1]

(c) Andrew wants to produce small 3D models of the company logo.

Explain how a 3D cutter could be used to produce the models.

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..... [2]

4 A zoo has an information point.

- Visitors use a menu to select information about animals.
- The menu includes 500 different animals.
- The information is provided only using high definition video with an audio track.

(a) State **one** input device that could be used for the information point.

..... [1]

(b) The output is shown on a monitor.

State **one** other output device that could be used for the information point.

..... [1]

(c) The video files are stored at the information point.

State **one** secondary storage device that could be used.

..... [1]

(d) The zoo decides to introduce Quick Response codes in different places in the zoo. These provide further information about the animals.

Describe how customers obtain the information from the Quick Response codes.

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..... [4]

(d) Give **two** benefits of using sensors to monitor the manufacture of plastic pipes.

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- 2
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- [2]

8 Explain how an instruction is fetched in a Von Neumann model computer.

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- [6]

- 2 The Von Neumann model for a computer system uses several components in the fetch-execute cycle. One component that is used is the Control Unit (CU).

Identify **four** other components that are used in the Von Neumann model for a computer system.

- 1
- 2
- 3
- 4

[4]

- 6 Touch screen technologies can be described as resistive or capacitive.

Six statements are given about resistive and capacitive technology.

Tick (✓) to show if the statement applies to **Resistive** or **Capacitive** technology.

Statement	Resistive (✓)	Capacitive (✓)
This touch screen has multi-touch capabilities		
This touch screen cannot be used whilst wearing gloves		
This touch screen is made up of two layers with a small space in between		
This touch screen uses the electrical properties of the human body		
This touch screen is normally cheaper to manufacture		
This touch screen has a quicker response time		

[6]

- 7 Gerald uses a keyboard to enter a website address into the address bar of his browser.

(a) Describe how Gerald's key presses on his keyboard are processed by the computer.

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[4]

11 Robert has a mobile device that uses RAM, ROM and an SSD.

(a) State what the RAM, ROM and SSD are used for.

RAM

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ROM

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SSD

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[3]

(b) Give **two** reasons why an SSD, rather than a HDD, is used in the mobile device.

Reason 1

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Reason 2

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[2]

0478/13 – NOVEMBER 2019

1 A library has a system that allows customers to check out the books that they want to borrow.

Each book has a barcode that can be used to identify the book.

(a) (i) Identify **two** input devices that may be used in the library's system.

Input device 1

Input device 2

[2]

(ii) Identify **two** storage devices that may be used in the library's system.

Storage device 1

Storage device 2

[2]

(iii) Identify **two** output devices that may be used in the library's system.

Output device 1

Output device 2

[2]

- (d)** The library often holds events that introduce new authors.

At the events, the library has a Liquid Crystal Display (LCD) screen that displays data, including an image and information about the author.

Describe how an LCD screen operates to display this data.

[5]

- (c) A sensor and a microprocessor are used to monitor the pH of the cleaning products. The system records each reading that is taken. If the reading is greater than 7 a warning message is displayed on a monitor.

Explain how the sensor and microprocessor are used in the system.

[6]

7 The Von Neumann model for a computer system has several components that are used in the fetch-execute cycle.

(a) One component is main memory.

(i) Describe what is meant by main memory and how it is used in the Von Neumann model for a computer system.

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..... [3]

(ii) State **two** other components in the Von Neumann model for a computer system.

1

2 [2]

- 1 The Von Neumann model for a computer system uses components, such as registers and buses, in the fetch-execute cycle.

(a) Draw a line to connect each component to its correct description.

Component	Description
Control Bus	Increments to point to the address of the next instruction to be fetched
Program Counter (PC)	Holds the result of a calculation. It is located within the Arithmetic Logic Unit (ALU)
Memory Data Register (MDR)	Carries signals to synchronise the fetch-execute cycle
Accumulator (ACC)	Temporary storage between the Central Processing Unit (CPU) and primary memory

[3]

- (b) State **two** buses, other than the control bus, used in the Von Neumann model for a computer system.

1

2

[2]

- 3 Priya studies music at school. She is buying a new computer to complete her school work at home.

- (a) Priya has a choice between an internal Hard Disk Drive (HDD) and an internal Solid State Drive (SSD) to store data.

- (i) Give **one** similarity between an HDD and an SSD.

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..... [1]

(ii) Explain **three** differences between an HDD and an SSD.

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[3]

(b) Priya needs to transfer files between the school and her home computer.

Identify **one** off-line storage device she could use to transport the files.

..... [1]

(c) Priya is using sound editing software to record and edit different music tracks.

(i) Identify **two** input devices she would use for this task.

Device 1

Device 2

[2]

(ii) Identify **two** output devices she would use for this task.

Device 1

Device 2

[2]

- 7 A room has an automatic lighting system. Electric lights are automatically turned on when a person enters the room and the natural light level in the room is 10 or less.

Explain how sensors and a microprocessor are used to control the electric lights in the room.

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- 1 A Von Neumann model for a computer system has a central processing unit (CPU) that makes use of registers.

(a) Identify **three** registers that may be used.

Register 1

Register 2

Register 3 [3]

(b) The CPU is responsible for processing instructions.

One stage of processing instructions is the decode stage.

(i) Identify the **two other** stages of processing instructions.

Stage 1

Stage 2 [2]

(ii) Identify the component of the CPU that is responsible for decoding instructions.

..... [1]

6 Six statements are given about touch screen technology.

Tick (✓) to show if the statement applies to **Capacitive** or **Resistive** touch screen technology.

Statement	Capacitive (✓)	Resistive (✓)
Needs pressure to be applied to create a circuit		
May not register a touch if the user is wearing gloves		
More commonly used in smartphones		
More responsive to a touch		
Needs an electrical field to be changed to register a touch		
Cheaper to manufacture		

[6]

- (b) State **two** benefits and **one** drawback of Leonard using a laser printer, instead of an inkjet printer, to print the letters.

Benefit 1

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Benefit 2

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Drawback

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[3]

9 (a) Six statements are given about storage devices.

Tick (✓) to show if the statement applies to hard disk drive (**HDD**) storage or solid state drive (**SSD**) storage.

Some statements can apply to both.

Statement	HDD (✓)	SSD (✓)
It has a limited number of read/write cycles		
It uses magnetic properties to store data		
It has moving parts		
It is non-volatile storage		
It can be used as an external storage device to back up data		
It uses flash memory to store data		

[6]

(b) Optical storage is another type of storage.

Give **two** examples of optical storage.

Example 1

Example 2

[2]

2 (a) Six hardware devices are shown.

Tick (✓) to show if each hardware device is an **Input**, **Output** or **Storage** device.

Hardware device	Input (✓)	Output (✓)	Storage (✓)
Solid state drive (SSD)			
Sensor			
Headphones			
Microphone			
USB flash drive			
Actuator			

[6]

(b) Genevieve writes a paragraph about a barcode reader.

Using the list given, complete the paragraph. Not all terms in the list need to be used.

- actuators
- binary
- black
- input
- microprocessors
- output
- sensors
- storage
- white

A barcode reader is an device. It shines a light at the barcode and the light is reflected back. The bars in the barcode reflect less light than the bars.

..... are used to capture the amount of reflected light and the different reflections are converted to values.

[5]

5 (a) A clothing shop uses a barcode reader at the checkout.

The checkout is linked to a stock control system. The system monitors stock levels and automatically keeps them above a minimum level.

Explain how the stock control system automatically keeps the stock levels above a minimum level.

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..... [4]

(b) The software for the stock control system is stored on a central computer. The computer uses random access memory (RAM), read only memory (ROM) and a hard disk drive (HDD).

The computer is a Von Neumann model computer system with a central processing unit (CPU).

(i) State the purpose of the RAM, ROM and HDD in the central computer.

RAM

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ROM

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HDD

..... [3]

(ii) Identify **four** components that are part of the CPU.

Component 1

Component 2

Component 3

Component 4 [4]

8 Six statements are given about printers.

Tick (✓) to show whether the statement applies to a **3D** printer, an **Inkjet** printer or a **Laser** printer.

Some statements apply to more than one printer.

Statement	3D (✓)	Inkjet (✓)	Laser (✓)
Uses a moving print head			
Uses liquid ink			
Produces output using materials such as plastic and resin			
Uses piezoelectric or thermal technology			
Uses a rotating drum to transfer the image to the paper			
Uses layer upon layer of material to create the output			

[6]

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1 Six devices are shown.

Tick (✓) to show if each device is an **Input**, **Output** or **Storage** device.

Device	Input (✓)	Output (✓)	Storage (✓)
Keyboard			
Sensor			
3D cutter			
2D scanner			
Microphone			
Hard disk drive (HDD)			

[6]

- (b)** Each ticket also has a QR code. The QR code is scanned at the entrance to the venue.

When a person enters, a count is incremented to show how many people have entered the venue.

[7]

- (b)** The computer has a central processing unit (CPU).

..... [1]

- (ii) Identify the register built into the arithmetic logic unit (ALU).

(iii) Four statements about a Von Neumann model for a computer system are shown.

Tick (✓) to show if each statement is **True** or **False**.

Statement	True (✓)	False (✓)
Data and instructions are stored in the same memory unit		
The control unit manages operations within the CPU		
Data and instructions can be fetched into the CPU at the same time		
The control unit is responsible for decoding an instruction		

[4]

8 Matthew is buying a new television with a display that uses LED technology.

(a) Explain what is meant by LED technology.

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..... [3]

(b) State **three** benefits of LED technology.

Benefit 1

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Benefit 2

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Benefit 3

..... [3]

(c) Identify **one other** technology that could have been used for the display.

..... [1]

(c) Tina uses a microphone to record a welcome message for her website.

(i) State whether the microphone is an **input** or **output** device.

..... [1]

(b) Alessandro uses an SSD to store his data.

Describe what is meant by an SSD and how it operates to store data.

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..... [4]

(c) Alessandro also uses off-line storage to store his data.

Three examples of off-line storage are Blu-ray, CD and DVD.

Six statements are given about off-line storage.

Tick (✓) to show if each statement applies to **Blu-ray**, **CD**, or **DVD**.

Some statements apply to more than one example of off-line storage.

Statement	Blu-ray (✓)	CD (✓)	DVD (✓)
A type of optical storage			
Has the largest storage capacity			
Can be dual layer			
Read using a red laser			
Has the smallest storage capacity			
Stores data in a spiral track			

[6]

5 Tammy is buying a new computer that has an LED display.

(a) Five statements about LED displays are given.

Tick (✓) to show if each statement is **True** or **False**.

Statement	True (✓)	False (✓)
It is a flat panel display		
It creates images using red, green and blue diodes		
It is not very energy efficient and gives off heat		
It can be used in mobile devices such as smartphones and tablets		
It is a front-lit display		

[5]

(b) Tammy connects the computer to her home network. The computer has a MAC address and an IP address.

A paragraph is given about MAC addresses and IP addresses.

Complete the paragraph using the list of terms given. Not all terms need to be used.

- compiled
- computer
- control
- dynamic
- identify
- packet
- principal
- protocol
- similar
- unique

A MAC address is a media access address.

A network device has a MAC address that
can help the device in the network. An IP address
is an Internet address. An IP address can be static or
.....

[5]

1 Five hardware devices are given.

Tick (✓) to show if each device is an **Input**, **Output** or **Storage** device.

Device	Input (✓)	Output (✓)	Storage (✓)
Solid state drive (SSD)			
Headphones			
2D cutter			
LCD projector			
Microphone			

[5]

2 Paige has a computer that has a central processing unit (CPU) based on the Von Neumann model for a computer system.

(a) Identify the component within the CPU that controls the flow of data.

..... [1]

(b) Identify the component within the CPU where calculations are carried out.

..... [1]

(c) Identify the component within the CPU that stores the address of the next instruction to be processed.

..... [1]

(d) Identify the register within the CPU that holds an instruction that has been fetched from memory.

..... [1]

(e) Identify the register within the CPU that holds data that has been fetched from memory.

..... [1]

8 Edith is buying a new computer monitor that displays images using LCD technology.

(a) Explain what is meant by LCD technology.

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..... [3]

(b) State **three** benefits of LCD technology.

Benefit 1

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Benefit 2

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Benefit 3

..... [3]

9 Elle uses both CDs and DVDs to store her school projects.

(a) Give **three** similarities between a CD and a DVD.

1

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..... [3]

(b) State **one** difference between a CD and a DVD.

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..... [1]

- The system for the game uses sensors and a microprocessor to spray water at a player as they run past each sensor.

[6]

- Five** statements about ISPs are given.

Statement	True (✓)	False (✓)
Provides access to the Internet for customers		
Can determine the maximum bandwidth available for customers		
Monitors the volume of data downloaded by customers		
Can provide an IP address for the customer		
Stores the content for all web pages available on the Internet		

©

- 3 The given table shows the name or description of four devices. The table is incomplete.

Complete the missing device names and descriptions.

Device name	Description
.....	Uses either thermal bubble or piezoelectric technology
Actuator	
.....	Uses thousands of tiny mirrors that can move very quickly to create an image
Mouse	

[4]

4 A supermarket sells many products. Each product has a barcode.

(a) Explain how the barcode is read at the supermarket checkout and how the price of the product is found.

[6]

(b) The supermarket stores data using a Solid State Drive (SSD).

(i) Explain how an SSD stores data.

[3]

(ii) One advantage of an SSD rather than a Hard Disk Drive (HDD) is that it has no moving parts, so it is more durable.

State **one** other advantage of the supermarket using SSD rather than HDD.

..... [1]

8 The Von Neumann model, for a computer system, uses the stored program concept.

(a) Describe what is meant by the stored program concept.

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..... [2]

(b) The fetch-execute cycle of a Von Neumann model, for a computer system, uses registers and buses.

(i) Describe the role of the Program Counter.

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..... [2]

(ii) Describe the role of the Control Bus.

.....
.....
..... [2]

(b) Benedict's computer is also assigned a Media Access Control (MAC) address.

(i) Identify **one** similarity between an IP address and a MAC address.

.....
 [1]

(ii) Identify **two** differences between an IP address and a MAC address.

Difference 1

 Difference 2

 [2]

2 Julia inputs personal data into her computer.

She stores three copies of the data using a hard disk drive (HDD), a solid state drive (SSD) and a USB flash memory drive.

(a) Identify **three** devices Julia can use to input personal data into her computer.

Device 1
 Device 2
 Device 3 [3]

(b) Six statements are shown about HDDs, SSDs and USB flash memory drives.

Tick (✓) to show which statements apply to each type of storage. Some statements can apply to more than one type of storage.

Statement	HDD (✓)	SSD (✓)	USB flash memory drive (✓)
it has no moving parts			
it is non-volatile			
it can use NAND gates to store data			
it uses magnetic properties to store data			
it has the smallest physical size			
it has the slowest read/write speeds			

[6]

- 5 Jamelia has a greenhouse that she uses to grow fruit and vegetables. She needs to make sure the temperature in the greenhouse stays between 25°C and 30°C (inclusive).

A system that has a temperature sensor and a microprocessor is used to maintain the temperature in the greenhouse. The system will:

- open a window and turn a heater off if it gets too hot
- close a window and turn a heater on if it gets too cold.

Describe how the system uses the temperature sensor and the microprocessor to maintain the temperature in the greenhouse.

[8]

2 Data storage can be magnetic, solid state or optical.

(a) Six statements are given about data storage.

Tick (✓) to show if the statement applies to magnetic, solid state or optical storage. Some statements may apply to more than one type of storage.

Statement	Magnetic (✓)	Solid state (✓)	Optical (✓)
no moving parts are used to store data			
pits and lands are used to store data			
data is stored on platters			
flash memory is used to store data			
parts are rotated to store data			
data can be stored permanently			

[6]

(b) (i) Give **one** example of magnetic storage.

..... [1]

(ii) Give **one** example of optical storage.

..... [1]

(iii) Identify which type of storage would be the most suitable for use in a web server and justify your choice.

Type of storage

Justification

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[3]

(c) Describe the operation of USB flash memory and how it stores data.

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..... [3]

- 5 A security light system is used by a factory. The light only comes on when it is dark and when movement is detected. The light will stay on for 1 minute before switching off.

Sensors and a microprocessor are used to control the security light system.

- (a) Identify **two** sensors that would be used in the security light system.

Sensor 1

Sensor 2

[2]

- (b)** Describe how the sensors and the microprocessor control the security light system.

[8]

8 A keyboard is a type of input device that can be used to enter data into a computer.

Complete the paragraph that describes one method of operation for a keyboard, using the most appropriate terms from the given list. **Not** all terms in the list need to be used.

- Binary
- Breaks
- Calculated
- Character
- Circuit
- Current
- Information
- Network
- Press
- Processor
- Signal
- Switch

A keyboard has a key matrix underneath the keys. When a key is pressed, it presses a that completes a This allows to flow. The location of the key pressed is The location of the key pressed is compared to a map to find the value for the key that has been pressed.

[6]

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(b) Greta stores data on several off-line storage devices, including an external hard disk drive (HDD), a USB flash memory drive and a compact disc (CD).

(i) Identify the type of storage for each device.

External HDD

USB flash memory drive

CD

[3]

(ii) Describe the operation of a HDD and how it stores data.

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..... [4]

3 An optical mouse is a type of input device that can be used to input data into a computer system.

(a) Complete the paragraph about the operation of an optical mouse, using the most appropriate terms from the given list. **Not** all terms need to be used.

- Ball
- Battery
- LCD
- LED
- Lens
- Magnifies
- Matrix
- Microswitch
- Photoelectric
- Photographic
- Reduces
- USB

An optical mouse shines an from the bottom of the mouse onto a surface. Light bounces straight back from the surface into a cell. This has a that the reflected light to allow detection of smaller movements. When a button on the mouse is clicked, a is pressed. A connection is used to carry the data to the computer.

[6]

(b) Identify **two** other input devices that can be used to enter data into a computer.

1

2

[2]

- 9 An underground car park has a system that checks the height of vehicles. A vehicle can be no higher than 1.8 metres to enter the car park.

The system also counts the number of vehicles that have entered the car park, so that it can display how many parking spaces are still available.

Each parking space has a red and a green light above it. If a car is parked in the parking space only the red light is on, otherwise only the green light is on.

Sensors and a microprocessor are used to control the system.

- (a)** Complete the table to identify a suitable sensor for each part of the system.

Task	Sensor
check if a vehicle is too high	
count the vehicles entering the car park	
check if a vehicle is parked in a parking space	

[3]

- (b)** Describe how the sensor and the microprocessor are used to display the red or green light above the parking space.

[6]

- 10 Several components are involved in processing an instruction in a Von Neumann model for a computer system.

Three of these components are the arithmetic logic unit (ALU), control unit (CU) and random access memory (RAM).

- (a) Six statements are given about the components.

Tick (✓) to show if each statement applies to the ALU, CU or the RAM. Some statements may apply to more than one component.

Statement	ALU (✓)	CU (✓)	RAM (✓)
stores data and instructions before they enter the central processing unit (CPU)			
contains a register called the accumulator			
manages the transmission of data and instructions to the correct components			
contained within the CPU			
uses the data bus to send data into or out of the CPU			
carries out calculations on data			

[6]

- (b) The accumulator is a register that is part of the Von Neumann model.

Give **two** other registers that are part of the Von Neumann model.

1

2

[2]

2 Magda has a mobile telephone.

She uses the touch screen on her telephone to send emails to her customers. The touch screen breaks, stopping Magda from using it to type her emails.

- (a) Identify **one** other input device that would be built into the mobile telephone that Magda could use to send an email to her customers.

..... [1]

- (b) The touch screen operates by using the conductive properties of the object that is used to touch the screen.

State whether the touch screen is a resistive, capacitive or infra-red touch screen.

..... [1]

- (c) Magda is listening to music on her mobile telephone when she receives a telephone call. A signal is sent within the telephone to stop the music and output that a call has been received.

Give the name of this type of signal.

..... [1]

4

- (c) Georgia uses a digital camera. The digital camera takes a photograph that is then converted into a digital image.

Complete the paragraph about the operation of a digital camera, using the most appropriate terms from the list. **Not** all terms in the list need to be used.

- analogue-to-digital
- binary
- charge-coupled
- digital-to-analogue
- lens
- light
- mirror
- pixel
- reflection
- sensor
- storage

When Georgia pushes the button to take a photograph, an aperture opens at the front of

the camera to allow to stream in through

the This is captured by a sensor called a

..... device. The

converter then converts each into a digital value.

[5]

6 Six statements are given about the role of components in the Central Processing Unit (CPU).

- (a) Tick (✓) to show whether each statement applies to the Memory Address Register (MAR), Memory Data Register (MDR) or Program Counter (PC).

Some statements may apply to more than one component.

Statement	MAR (✓)	MDR (✓)	PC (✓)
it is a register in the CPU			
it holds the address of the next instruction to be processed			
it holds the address of the data that is about to be fetched from memory			
it holds the data that has been fetched from memory			
it receives signals from the control unit			
it uses the address bus to send an address to another component			

[6]

- (b) Identify the component in the CPU that carries out calculations.

..... [1]

- 7 (a) Tick (✓) **one** box to identify if an internal Solid State Drive (SSD) is an example of primary, secondary or off-line storage. Justify your choice.

Tick (✓)

☐

Primary

☐

Secondary

☐

Off-line

Justification

.....

.....

.....

[3]

[4]

(a) A sensor is used in each of the given tasks.

Each sensor given must be different.

[3]

This image shows a full page of white paper with ten horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and extend across the entire width of the page. There is no text or other markings on the paper.

- 2 A train company wants to install a self-service ticket machine system for its train stations. When the customer has purchased their tickets, the machine will provide a paper ticket.

(a) **One** output device that is used in the ticket machine is a display screen.

Identify **one** other output device that is used in the ticket machine system.

..... [1]

(b) The train company does **not** want users to use a keyboard or a mouse to enter their data, when buying a ticket. The company is worried that they may be stolen or get too dirty.

Identify **one** other input device that would be suitable for use in the ticket machine system, to allow users to enter their data.

..... [1]

- 4 The paragraph explains the operation of different touch screen technologies.

Complete the paragraph using the list of terms. **Not** all terms in the list need to be used.

- capacitive
- change
- circuit
- conductive
- coordinates
- grid
- heat
- infra-red
- insulating
- light
- manufacture
- pressure
- resistive

In touch screen technology, an electrostatic field is present on the surface of the touch screen. The properties of a user cause a in the field. The of the user's touch can be calculated.

In touch screen technology, a user pushes the top layer of the screen and makes it connect with the bottom layer to complete a

This type of touch screen is cheaper to

[7]

7 Five statements are given about devices.

Tick (✓) to show if each statement applies to a 3D scanner, barcode reader or a Quick Response (QR) code reader. Some statements may apply to more than **one** type of device.

Statement	3D scanner (✓)	Barcode reader (✓)	QR code reader (✓)
uses position and alignment markers for orientation when scanning			
scans the shape and appearance of an object			
uses reflected light from a laser to convert a black-and-white pattern into binary			
can often be built into an Electronic Point Of Sale (EPOS) terminal, for example, a supermarket checkout			
it is an example of an input device			

[5]

8 An electronic game has **three** square mats that are coloured red, green and blue.

The player will see a colour displayed on a screen and has 1 second to hit the mat that matches the colour. If the player hits the correct mat, within 1 second, a counter is incremented. When a player hits an incorrect mat, the game ends.

The game uses sensors and a microprocessor to determine if the player hits the correct mat within 1 second.

Explain how the game uses sensors and a microprocessor to count the number of times a player hits a correct mat within 1 second.

This image shows a full page of a worksheet designed for handwriting practice. It features ten evenly spaced, horizontal dashed lines across the entire width of the page. The background is plain white, and there are no margins, text, or other markings present.

2

(d) Sensors are used at the entrance to count the number of people entering the stadium.

- (i) Identify **two** sensors that could be used to count the number of people entering the stadium.

Sensor 1

Sensor 2

[2]

- (ii) Tick (✓) **one** box to show if a sensor is an example of an input device, storage device or output device.

Device	Tick (✓)
input	
storage	
output	

[1]

- 5 In a Von Neumann model for a computer system, a Central Processing Unit (CPU) contains a number of different components.

The table contains the name of a component or a description of their role in the fetch-execute cycle.

Complete the table with the missing component names and descriptions.

Component name	Description
Memory Address Register (MAR)	
Program Counter (PC)	
.....	This is a register that is built into the arithmetic logic unit. It temporarily holds the result of a calculation.
.....	This is a register that holds data or an instruction that has been fetched from memory.
Control Unit (CU)	
.....	This carries addresses around the CPU.

[6]

4 Pradeep uses his personal computer to complete work at home.

(a) Pradeep uses a mouse and a keyboard to control the computer.

(i) Complete the descriptions of the principles of operation of a mouse and a keyboard using the most appropriate terms from the list. **Not** all terms in the list need to be used.

- absorbs
- ball
- biometric
- circuit
- colour
- digital
- direction
- Light-Emitting Diode (LED)
- Liquid Crystal Display (LCD)
- reflects
- speed
- switch
- transparency

An optical mouse shines a red light from a underneath the mouse. The light back from a surface through a lens in the mouse and is converted to a value. This value is transmitted to the computer. The computer then determines the and of the movement.

When the user presses a key on a keyboard, the key pushes the on the circuit board. This completes a Signals are sent to the computer. The computer uses the data to calculate which key was pressed.

[6]

(ii) Identify **two** other input devices Pradeep could use with his personal computer.

Input device 1

Input device 2

[2]

(b) Pradeep uses a projector attached to the ceiling at his home to watch high-definition (HD) films.

The projector has broken. He wants to buy a replacement. He needs to choose between an LCD projector and a Digital Light Projector (DLP).

Explain why an LCD projector would be more appropriate for Pradeep.

.....

.....

.....

.....

.....

..... [3]

(c) Pradeep stores his collection of films and his work files on his personal computer.

Pradeep wants to save a copy of all his films and files onto a single storage device.

Identify and justify an appropriate storage device to store the copies.

Storage device

Justification

.....

.....

.....

.....

.....

..... [4]

2 A computer is designed using the Von Neumann model for a computer system.

The computer has a central processing unit (CPU).

(a) Data is fetched from primary storage into the CPU to be processed.

(i) State the name of the primary storage from where data is fetched.

..... [1]

(ii) The CPU performs a cycle to process data. Fetch is the first stage in this cycle.

State the names of the second and third stages in the cycle.

Second stage

Third stage [2]

(iii) Identify **two** components within the CPU that are used in the fetch stage of the cycle.

Component 1

Component 2 [2]

3 Three types of storage media are magnetic, optical and solid state.

(a) One example of solid-state storage is a Solid State Drive (SSD).

Identify **one** other example of solid-state storage.

..... [1]

(b) Optical storage uses a laser to store and read data from a disk.

Explain how the laser is used to store and read data from the disk.

.....
.....
.....
.....
.....
..... [3]

(c) A business is creating a new mobile device that has an SSD as secondary storage.

- (i)** Give **three** reasons why an SSD is the most suitable secondary storage for their mobile device.

Reason 1

.....

Reason 2

.....

Reason 3

.....

[3]

- (ii)** Identify **two** examples of software that can be stored on the SSD.

Example 1

Example 2

[2]

10 Many devices have a Media Access Control (MAC) address.

Give **three** features of a MAC address.

Feature 1

.....

Feature 2

.....

Feature 3

.....

[3]

11 (a) The paragraph describes the process of printing a document using an inkjet printer.

Complete the paragraph using the most appropriate terms from the list. **Not** all of the terms in the list need to be used.

- binary
- buffer
- drum
- information
- interrupt
- laser
- liquid
- nozzles
- operating system
- powder
- thermal bubble
- toner

Data is sent from the computer to the printer. The data is held in a print

..... that is temporary storage until the data is processed to be printed.

Inkjet printers operate by having a print head that moves

..... side to side across the page. These spray ink droplets onto the page. These ink droplets can be created using piezoelectric or technology.

If the paper jams in the printing process, the printing stops and an

..... is sent to the computer.

[5]

(b) A printer is one example of an output device.

Give **three** other examples of output devices.

Example 1

Example 2

Example 3

[3]

(c) Give **three** examples of input devices.

Example 1

Example 2

Example 3

[3]

7 Cassie stores data for her business every day. She stores the data using optical data storage.

(a) Identify **three** examples of optical data storage.

Example 1

Example 2

Example 3

[3]

(b) **Six** statements are given about the operation of three different types of storage.

Tick (✓) to show which statements apply to each type of storage. Some statements may apply to more than **one** type of storage.

Statement	Type of storage		
	Magnetic (✓)	Optical (✓)	Solid state (✓)
this storage has no moving parts			
this storage uses a laser to read and write data			
this storage uses a read/write head			
this storage burns pits onto a reflective surface			
this storage uses NAND and NOR technology			
this storage stores data in tracks and sectors			

[6]

- 1 (a) The table lists **five** devices.

Tick (✓) to show if each device is an example of an input device, an output device or a storage device.

Device	Input (✓)	Output (✓)	Storage (✓)
keyboard			
microphone			
headphones			
hard disk drive (HDD)			
actuator			

[5]

- (b) There are three different types of storage device. One type is magnetic.

State the other **two** types of storage device.

Type 1

Type 2

[2]

- 2 (b) The display screen used above each seat is a Light-Emitting Diode (LED) display screen.

Give **two** reasons why an LED display screen has been used.

Reason 1

.....

Reason 2

.....

[2]

The computer has a central processing unit (CPU).

(a) Explain the purpose of the CPU.

Identify the **three** stages of the cycle performed by the CPU in your answer.

..... [5]

(b) The memory address register (MAR) is one component built into the CPU.

(i) Identify **four** other components that are built into the CPU.

Component 1

Component 2

Component 3

Component 4

[4]

ii) Describe the purpose of the MAR in the CPU.

..... [2]

[2]

(c) Harriet's computer has both primary and secondary storage.

(i) Explain what is meant by primary storage.

Give **three** examples of what is stored in primary storage in your answer.

..... 17

(ii) The secondary storage in Harriet's computer is damaged, so it is removed for repair. This means the computer will no longer work.

Explain why the secondary storage is needed for the computer to work.

.....

.....

.....

..... [2

- (c) Frida scans a text document into her computer so that she can store a digital version. She uses a 2D scanner to do this.

Describe how the text document is scanned by the 2D scanner to create the digital version.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Random access memory (RAM) is an example of primary storage.

Give **three** examples of data that is commonly stored in RAM.

- 1
- 2
- 3 [3]

(c) Describe the purpose of secondary storage.

.....
.....
..... [2]

(c) The photographer decides to purchase a solid-state storage device to back up their images.

Complete the description of solid-state storage.

Use the terms from the list.

Some of the terms in the list will **not** be used. You should only use a term once.

binary denary electrons grid neutrons
non-volatile RAM star transistors virtual volatile

Solid-state storage is This means that the data is **not** lost when the power is turned off.

Solid-state storage is made of that are laid out in a

Gates are used to control the flow of the through the transistors. This changes the data in the transistors from 1 to 0, or from 0 to 1.

[4]

3 Secondary storage devices are used to store data in a computer.

(a) Circle **three** components that are secondary storage devices.

central processing unit (CPU)

compact disk (CD)

hard disk drive (HDD)

random access memory (RAM)

read only memory (ROM)

register

sensor

solid-state drive (SSD)

[3]

(b) Tick (✓) **one** box to show which statement about secondary storage is correct.

A It is directly accessed by the CPU.

☐

B It is magnetic storage only.

☐

C It is used to permanently store software and data files.

☐

D It is volatile.

☐

[1]

- 5** A farm has an automated drinking system for its animals. The drinking system has a water bowl that contains the water. When the water bowl is empty, it is automatically refilled.

The system uses a sensor and a microprocessor.

- (a)** Identify the most appropriate sensor for this system.

..... [1]

- (b) Describe how the sensor and the microprocessor are used to automatically refill the water bowl.

[6]

6 A user wants to connect their computer to a network.

(a) (i) Identify the component in the computer that is needed to access a network.

..... [1]

(ii) Identify the type of address that is allocated to the component by the manufacturer, which is used to uniquely identify the device.

..... [1]

(b) A dynamic internet protocol (IP) address is allocated to the computer when it is connected to the network.

(i) Identify the device on the network that can connect multiple devices and automatically assign them an IP address.

..... [1]

(ii) Describe what is meant by a dynamic IP address.

.....
.....
.....
.....
.....
..... [3]

- 1 Output devices are used to output data from a computer.

Circle **three** devices that are output devices.

actuator

digital versatile disk (DVD)

keyboard

microphone

mouse

printer

scanner

sensor

solid-state drive (SSD)

speaker

[3]

- (b) The photographs are also transmitted across a network to cloud storage. A device on the network forwards the data towards its correct destination.

- (i) State the name of this device.

..... [1]

- (ii) Describe what is meant by cloud storage.

.....
.....
.....
..... [2]

- (iii) Give **one** disadvantage of storing the photographs in cloud storage instead of storing them locally.

.....
..... [1]

8 A computer is connected to a network and assigned an IPv4 address.

(a) Tick (✓) **one** box to show which device would assign the IPv4 address to the computer.

A Domain name server (DNS)

☐

B Network interface card (NIC)

☐

C Router

☐

D Web server

☐

[1]

(b) Describe the characteristics of an IPv4 address.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 2 A library has a self-checkout system that allows customers to register books that they want to borrow.

The self-checkout system has a central processing unit (CPU).

The CPU has two cores.

- (a) State the purpose of a core in the CPU.

.....
..... [1]

- (b) The CPU is replaced with one that has four cores.

Explain the effect this has on the performance of the self-checkout system.

.....
.....
.....
..... [2]

- (c) The CPU contains registers and buses.

- (i) Describe the role of a register in the CPU.

.....
.....
.....
..... [2]

- (ii) Identify **one** bus that can be found in the CPU and explain its purpose in the fetch–decode–execute cycle.

Bus

Purpose

.....
.....
.....
..... [3]

(d) The self-checkout system allows the user to input their library membership number.

Give **two** appropriate input devices that would allow the user to do this.

1

2 [2]

(e) The self-checkout system uses a monitor to display information about the book.

Users who are blind also need to use the self-checkout system.

Give an appropriate output device that would allow a blind user to be given this information.

..... [1]

(f) The self-checkout system uses two types of primary storage.

Circle **two** types of primary storage that would be used in the system.

compact disk (CD)

digital versatile disk (DVD)

hard disk drive (HDD)

random access memory (RAM)

read only memory (ROM)

universal serial bus (USB) flash memory drive

[2]

(g) The self-checkout system is linked to a stock control system that is updated every time a book is borrowed from the library.

A microprocessor is used in the stock control system to update the stock.

Explain the role of the microprocessor in this system and how it is used to update the stock when a book is borrowed.

.....

.....

.....

.....

.....

..... [3]

3 Five network terms or definitions are given in the table.

Complete the table by giving the missing term or definition.

Term	Definition
router	
.....	This address is assigned by the network and used to identify a device on a network.
network interface card (NIC)	
.....	This address is assigned by the manufacturer and is used to uniquely identify the device.
.....	This can be hardware or software based and filters traffic coming into and out of a network.

[5]



9 A device can be given an internet protocol (IP) address. This can be an IPv4 or IPv6.

(a) Give **one** similarity between IPv4 and IPv6.

.....
..... [1]

(b) Describe **two** differences between IPv4 and IPv6.

1
.....
.....
.....
2
.....
.....
..... [4]

(c) A web page is requested using an IP address.

(i) Identify the system that stores a database of uniform resource locators (URLs) and their corresponding IP addresses.

..... [1]

(ii) Identify the software that sends a request to the IP address to obtain the web page data.

..... [1]

10 A computer has pages A, B and C that are stored in RAM. Page D needs to be sent to the RAM but the RAM is full.

Page B is **not** needed immediately.

Explain how virtual memory can be used in this scenario.

.....
.....
.....
.....
.....

- 1 A student uses a computer and several hardware devices to complete his schoolwork.

The computer has a central processing unit (CPU).

- (a) The student uses a keyboard to complete his schoolwork.

Tick (✓) **one** box to show which type of device the keyboard is.

- | | | |
|---|---------|--------------------------|
| A | input | ☐ |
| B | memory | ☐ |
| C | output | ☐ |
| D | storage | ☐ |

[1]

- (b) The student uses a printer to print his schoolwork.

Tick (✓) **one** box to show which type of device the printer is.

- | | | |
|---|---------|--------------------------|
| A | input | ☐ |
| B | memory | ☐ |
| C | output | ☐ |
| D | storage | ☐ |

[1]

- (c) A component in the CPU sends signals to manage the fetch-decode-execute cycle.

State the name of this component.

..... [1]

- (d) The CPU has a clock speed of 2.4 Ghz.

Describe what is meant by a 2.4 Ghz clock speed.

.....
.....
.....
..... [2]

(e) The CPU contains registers including the memory data register (MDR).

(i) Describe the role of the MDR in the fetch-decode-execute cycle.

.....

.....

.....

..... [2]

(ii) Identify **three** other registers contained in the CPU.

1

2

3 [3]

2 (e) When a customer is leaving the car park they arrive at a barrier. The customer needs to insert their ticket into a system at the barrier. This system reads the ticket number then checks whether the parking cost has been paid for the car. The barrier is raised if it has been paid.

The system uses a microprocessor.

Describe the role of the microprocessor in the system and how it checks whether the parking cost has been paid.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

8 Storage can be described as being magnetic, solid-state or optical.

(a) Give **two** features of magnetic storage.

1

.....

2

.....

[2]

(b) Give **three** features of solid-state storage.

1

.....

2

.....

3

.....

[3]

(c) Give **one** example of each type of storage.

Magnetic

Solid-state

Optical

[3]

- 3 A user's computer has a central processing unit (CPU) that has a clock speed of 2 GHz.

She wants to change it to a CPU that has a clock speed of 3 GHz.

- (a) (i) State what is meant by clock speed.

.....
..... [1]

- (ii) Explain the effect this change will have on the performance of the CPU.

.....
.....
.....
..... [2]

- (b) The CPU contains a memory address register (MAR).

Describe the role of the MAR in the fetch–decode–execute cycle.

.....
.....
.....
..... [2]

- (c) The CPU has a list of all the machine code commands it can process.

State the name of this list of commands.

..... [1]

- 4 A washing machine is an example of an embedded system.

- (a) Give **two** characteristics of an embedded system.

1
.....
2
.....
..... [2]

13 Storage and memory are important components of a computer system.

(a) Primary storage is one type of storage in a computer system.

(i) Tick (✓) **one** box to show which is an example of primary storage.

A compact disk (CD)

☐

B hard disk drive (HDD)

☐

C random access memory (RAM)

☐

D solid-state drive (SSD)

☐

[1]

(ii) Give **one** characteristic of primary storage.

..... [1]

(b) Virtual memory can be created in a computer system.

Complete the description about virtual memory.

Use the terms from the list.

Some of the terms in the list will **not** be used. Some terms may be used more than once.

binary

hard disk drive (HDD)

hexadecimal

operating system

pages

random access memory (RAM)

read only memory (ROM)

sectors

software

tracks

virtual memory

Virtual memory is used when the is full. It is
created by partitioning the Data is divided into
..... that can be sent from
..... to the
..... to be temporarily stored until they are required.

[5]

1 A mobile telephone has built-in input and output devices.

(a) Give **two** examples of an input device that would be built into a mobile telephone.

1

2 [2]

(b) Give **one** example of an output device that would be built into a mobile telephone.

..... [1]

3 A computer has a central processing unit (CPU).

(a) Circle **three** components that are built into the CPU.

accumulator (ACC)

control unit (CU)

graphics card

hard disk drive (HDD)

motherboard

program counter (PC)

random access memory (RAM)

read only memory (ROM)

[3]

(b) The CPU has cache.

Explain the purpose of the cache.

.....

.....

.....

..... [2]

(c) The CPU has a component that regulates the number of fetch–decode–execute cycles the CPU can perform in a second.

State the name of this component.

..... [1]

(d) The CPU has a component that carries out all calculations and logical operations.

State the name of this component.

..... [1]

6 A company uses cloud storage to store its data.

(a) Tick (✓) **one** box to show which is **not** a characteristic of cloud storage.

- A

Data is accessed through a network

☐
- B

Data is stored locally

☐
- C

Data is stored remotely

☐
- D

Physical servers are used to store the data

☐

[1]

(b) Explain **two** advantages for the owners of the company of storing its data in cloud storage.

- 1

.....

.....

.....

.....
- 2

.....

.....

.....

.....

[4]

(c) Explain **one** disadvantage to employees of the company storing data in the cloud.

-

.....

.....

.....

[2]

8 Draw and annotate a diagram to represent the role of a router.

[4]

9 A computer has secondary storage.

(a) The table contains statements about secondary storage.

Complete the table by writing the type of secondary storage that applies to each statement. Some types of secondary storage may apply to more than one statement.

Type of secondary storage	Statement
.....	data is stored using pits and lands
.....	data is stored using control gates and floating gates
.....	data is stored using electromagnets
.....	data is stored using a laser
.....	data is stored on a platter that is divided into tracks and sectors

[5]



(b) Explain **two** differences between primary storage and secondary storage.

1
.....
.....
.....

2
.....
.....
.....